

Lab Experiment: Constructor and Destructor in OOP (C++)

Aim:

To write a C++ program that demonstrates the use of constructor and destructor in a class.

Theory:

- A constructor is a special member function of a class that is automatically invoked when an object is created.

* It has the same name as the class.

* It does not have a return type.

* Can be parameterized or default.

- A destructor is a special member function invoked automatically when an object goes out of scope or is explicitly deleted.

* It has the same name as the class but is preceded by a ~ symbol.

* It is used to release resources (like memory, files, connections).

* A class can have only one destructor, and it cannot be overloaded.

Program (C++):

```
#include <iostream>
using namespace std;

class Student {
    string name;
    int rollNo;

public:
    // Constructor
    Student(string n, int r) {
        name = n;
        rollNo = r;
        cout << "Constructor called for " << name << endl;
    }

    // Display function
    void display() {
        cout << "Name: " << name << ", Roll No: " << rollNo << endl;
    }

    // Destructor
    ~Student() {
        cout << "Destructor called for " << name << endl;
    }
};

int main() {
    // Creating objects
    Student s1("Alice", 101);
    Student s2("Bob", 102);

    cout << "\nDisplaying Student Details:" << endl;
    s1.display();
    s2.display();

    return 0;
}
```

Expected Output:

```
Constructor called for Alice  
Constructor called for Bob
```

```
Displaying Student Details:  
Name: Alice, Roll No: 101  
Name: Bob, Roll No: 102  
Destructor called for Bob  
Destructor called for Alice
```

Viva Questions & Answers:

Q1. What is a constructor in C++?

■ A constructor is a special member function that initializes objects when they are created.

Q2. Can constructors be overloaded?

■ Yes, multiple constructors with different parameter lists can exist (constructor overloading).

Q3. What is a destructor in C++?

■ A destructor is a special member function that destroys objects and releases resources when they go out of scope.

Q4. Can we overload a destructor in C++?

■ No, a class can have only one destructor, and it cannot be overloaded.

Q5. When is the destructor automatically called?

■ It is automatically called when an object goes out of scope or is explicitly deleted (for dynamically allocated objects).